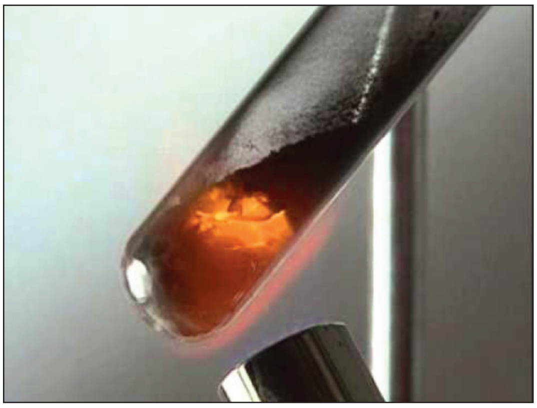
Examiner only

5. *(a)* A teacher showed her class how copper can be extracted from copper(II) oxide using charcoal.



$$2CuO(s) + C(s) \longrightarrow CO_2(g) + 2Cu(s)$$

In her experiment, the teacher heated 1.59 g of copper(II) oxide and 0.12 g of charcoal in a boiling tube. She continued to heat the mixture strongly until it had been glowing for 5 minutes.

She recorded the mass of the boiling tube and its contents before heating and then again after heating, once the carbon dioxide produced had been released from the tube.

Mass of boiling tube = 37.43 g
Mass of boiling tube and contents **before** heating = 39.14 g
Mass of boiling tube and contents **after** heating = 38.82 g

(i) The teacher had expected the reaction to produce 1.27 g of copper from the masses of copper(II) oxide and charcoal used. Use her results to show that the mass produced suggests a 109 % yield. [2]

(ii) Assuming the reactants had been heated at a high enough temperature for sufficient time, suggest **one** reason for the yield being larger than expected. [1]

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Examiner only

(b) Iron is extracted from iron(III) oxide, Fe_2O_3, inside the blast furnace using coke.

(i) Write a balanced **symbol** equation for **one** of the reactions that show the reduction of iron(III) oxide inside the furnace. [2]

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(ii) An iron ore contains 22 % by mass of iron(III) oxide. Calculate the maximum mass of iron that could be obtained from 5×10^5 tonnes of this ore. Give your answer in **standard form**. [3]

Mass = ... tonnes

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